CSCI 272 Formal Languages **Spring 2025 - Homework Exercise 4**

This homework assignment is worth 24 points total.

Solve the following problems to the best of your ability, writing your solutions by hand on white paper. Don't write too small, and use clear handwriting. You may use a pencil or pen, but not one that uses red ink.

You will need to submit your solutions both physically and digitally. The digital copy should be submitted to Moodle by Tuesday, April 22 by 23:59. The hard copy will be due on Wednesday, April 23 just before the start of your lecture section. The physical copy must be identical to what was submitted digitally.

Scanning Guidelines:

- Make a good quality scan. Phone pictures (CamScanner) are acceptable, as long as they are in focus and readable.
- Before submitting, verify that you can read your own scans.
- Scan the pages in the correct orientation (not rotated over 90 degrees.)
- You may submit your scans in a single .pdf file with the solutions presented in the same order as the questions, or collect them into a single archive file (.zip, .gzip, or .rar) with your files numbered in such a way that they appear in the right order.

---

1. (18 pts.) State which of the following three classes the following languages belong to and justify your answers. You may use reductions in your justifications.

- Decidable
- Recognizable, but not decidable (← Requires two proofs!)
- Unrecognizable

a) $F \cup \overline{A_{TM}}$, where F is an arbitrary finite language

b) $\overline{F_1} \cup F_2$, where F_1 and F_2 are both finite languages

c) { w ∈ {0, 1}* | w is the binary representation of a prime number }

d) $A_{\geq 3}$ = { <M> | M accepts at least three different strings }

e) { <M, w> | <M, w> is one of the first 500 elements (in alphabetical order) of $\overline{A_{TM}}$ }

f) EQ_{TM} = { <M_1, M_2> | M_1 and M_2 are TMs that accept exactly the same strings }

g) $\overline{EQ_{TM}}$

h) $A_{PRIME30}$ = { <M> | M accepts the binary representation of the first 30 prime numbers }

i) A_{INF} = { <M> | M is a TM that accepts an infinite number of strings }

2. (6 pts.) Find examples of the following languages which fit the given descriptions, and justify your answers.

a) A language L, such that both L and $\overline{L}$ are both decidable and infinite

b) A language L, such that both L and $\overline{L}$ are both unrecognizable

c) A decidable language D, and an unrecognizable language N, such that their union $D \cup N$ is decidable